2025 秋个人研究计划（Research Proposal）

重要提示：

1. *不要修改或删除已有内容或文本格式包括字体、字号、字色以及各类间距边距等；*
2. *正文部分英文: 300-400 词、中文: 450-600 字；*
3. *写作完毕后导出为 **PDF 格式**，用**班级+姓名**格式命名文件。*

基本信息 （Information）

班级（Class):B6

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学号（Student ID):20251800903

所在小组编号（Group No.):1 组

研究计划标题 （Proposed Research Title）:

中 文： 干旱地区生态恢复：矿区土壤改良、植被恢复与地下水补给

英 文 ： Ecological Restoration in Arid Regions: Soil Improvement, Vegetation Recovery, and Groundwater Recharge in Mining Areas

第一段：研究背景 （Introduction / Background）

写作指引： 此段对应大纲的第 1 部分。请介绍宏观背景，强调话题重要性，并引出核心文献及其贡献，最后自然地过渡到现有研究的局限性。

The impact of mining on the environment,especially in arid and semi-arid areas, is a major issue today. Coal mining not only damages the soil and vegetation but can also deplete groundwater resources, making ecological restoration more challenging. Many studies have focused on solving one issue at a time, such as improving soil or vegetation, but haven’t looked at how soil,vegetation, and groundwater can work together.This research will build on the methods used in the paper Varying Effects of Mining on Vegetation and Groundwater Storage in Inner Mongolia and combine them with the concept of “green mining” to explore how to simultaneously improve soil,restore vegetation, and recharge groundwater, so that these factors can c√ lement each other and help restore mining areas effectively.

第二段：问题陈述与研究空白 （Problem Statement / Research Gap）

写作指引： 此段对应大纲的第 2 部分。清晰地、有理有据地论证现有研究的不足（即“研究空白”），并强调填补此空白的必要性和价值。

Currently, most research on ecological restoration in mining areas focuses on one aspect, such as vegetation or groundwater, and overlooks the importance of soil quality. In arid and semi-arid areas, the relationships between soil, vegetation,and groundwater are very complex. While some studies have discussed the interactions between these factors, there is a lack of a comprehensive system to consider all three at once. This creates a gap in our ability to develop a unified approach to assess and guide ecological restoration in mining areas. Therefore, filling this gap is essential to create a framework that takes into account soil,vegetation, and groundwater together, which will help guide more effective and sustainable restoration efforts

第三段：研究方案与预期贡献 (Research Plan & Expected Contribution)

写作指引： 此段浓缩大纲的第 3、4、5 部分。用简洁、有力的语言，依次完成三件事：1. 正式提出你的研究问题/目标；2. 简要介绍你将采用的研究方法；3. 预告你研究的预期成果与意义。

The study will transplant and transform the IEI and GGRACE framework from the paper and apply it to a multi-mineral, multi-scale mine area data set (6-9mines). On this basis, engineering indicators such as soil improvement and mining scale will be introduced to evaluate the restoration effect and establish anintegrated evaluation system.It is expected that this study will establish a “soil-vegetation-groundwater” coupled system applicable to different types of minerals and propose a diagnostic and predictive model for ecological restoration effects. This research will provide mining engineers, policymakers, and green mine builders with specific evaluation tools and restoration strategies, promoting the standardization and scientific approach to mine ecological restoration and advancing the sustainable development of the mining industry.